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Critical Realism in Quantum Mechanics

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Abstract

In this talk, we will develop a philosophical lens to investigate classical materialism and the problem of realism in quantum mechanics, leading up to quantum relationalism, a critical realist approach to quantum foundations.

The part on classical materialism studies the contributions of Heraclitus, Plato, Spinoza, Leibniz, Berkeley, Hume, Kant, Frege, Russell, Mach, Bogdanov, Anscombe, Quine, Davidson, Putnam and BonJour, culminating in the Newtonian paradigm of physics.

In this backdrop, we interpret quantum mechanics via the Bohr-Heisenberg-Born-Dirac school of anti-realists; the Einstein-de Broglie-Schrödinger-Bohm school of realists; and the Planck-Lamb-Feynman school of pragmatists. We then interpret issues in quantum foundations via the modern Everett-DeWitt-Barbour-Wallace magical realist school, the Bayes-Gibbs-Ramsey-Fuchs epistemic school and finally, the Rovelli-Smolin-Crane critical realist school.

As a conclusion, we explore how quantum relationalism suggests holography in quantum gravity, exhibiting structural realism in the evolution of scientific theories.

Outline

1. Philosophy of philosophy
   - Traditional epistemology
   - Modern epistemology
   - Epistemic coherentism
2. Philosophy of science
   - Ontological positions in science
   - Scientific realism
   - Scientific anti-realism
   - Dualism
3. Classical materialism
   - Event ontology
   - Causal metaphysics
   - Object ontology
   - Logical positivism
   - Synthesis

Traditional philosophy

- Philosophy = $\underbrace{\text{Philos}}_{\text{Love}}$ + $\underbrace{\text{Sophia}}_{\text{Knowledge}}$ (Greek).
- **Socrates** (470 – 399 BC) added a *justificatory* dimension to true beliefs. This gradually gave rise to the traditional *tripartite analysis of knowledge*:

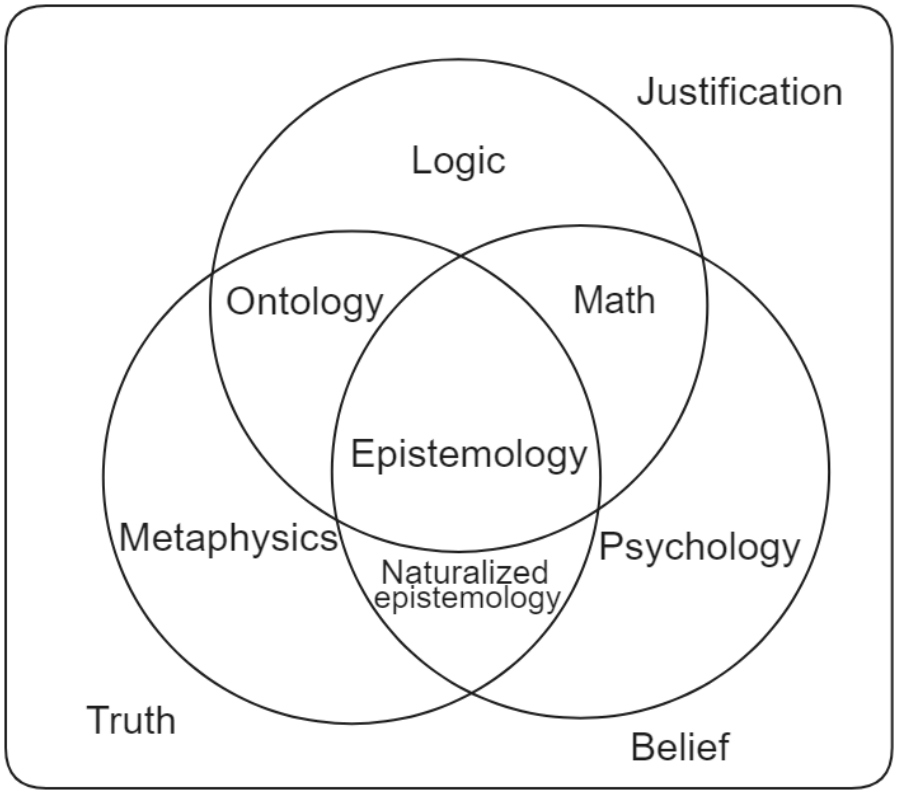


Figure: The traditional landscape of philosophy.

Modern epistemology

- **Edmund Gettier** (1927 - 2021) revolutionized epistemology by questioning the status of knowledge as justified-true-belief (JBT). One aspect of *Gettier problems* in the tripartite analysis is that **foundationalist inferences do not preserve the epistemic status of propositional attitude statements**:

  Let p be a proposition and J, B_S be justification and belief (with respect to an agent S) predicates. In an *internalist* theory of justification, J is with respect to the knowledge K_S of S.

  Suppose $J_S(p) \wedge B_S(p)$ but $\neg p$. Consider an auxiliary proposition q with the referent '*true*', which S in unware of, i.e. $\neg K_S(q)$. By disjunction introduction,

$$
\begin{aligned}
p \vee q &\implies J_S(p \vee q) \wedge (p \vee q) \wedge B_S(p \vee q) \\
\text{However, } J_S(p) \wedge \neg p \wedge B_S(p) &\implies \neg[J_S(p) \wedge p \wedge B_S(p)] \\
&\implies \neg K_S(p) \\
\neg K_S(p) \wedge \neg K_S(q) &\implies \neg K_S(p \vee q) \\
\therefore J_S(p \vee q) \wedge (p \vee q) \wedge B_S(p \vee q) &\not\Longrightarrow K_S(p \vee q)
\end{aligned}
$$

- According to **Alvin Goldman** (born 1938), S has no knowledge of $p \vee q$ in the above Gettier case as there is **no causal connection between the referents of the true clauses of the proposition and its belief in** S:

$$q \not\to B_S(q)$$

In Goldman's *causal theory of knowledge*, justification must therefore entail the existence of such *external* causal relationships,

$$J_S(p \vee q) \wedge \neg p \wedge q \implies q \to B_S(q)$$

- **Epistemic relativism** is the realization that not only is knowledge causal, but also *relative to the agent*, i.e., $K_S(p)$ involves the causal relationship between the referents of p and S itself,

$$K_S(p) \implies p \to S$$

In a relativist epistemology, when $p \not\to T$, we have $\neg K_T(p)$. However, if $p \to U$, the agent U potentially agrees with the agent S, i.e.,

$$K_S(p) \wedge (p \to U) \not\Rightarrow \neg K_U(p)$$

In **monistic relativism** (e.g. topos theory), a rational agent U can, in principle, perform rational procedures to agree with the contents of p. This is not necessary in **pluralistic relativism**.

Epistemic coherentism

- The **trilemma of linear justification** is as follows:
  1. Linear justification is either infinite (*infinite regress*), or,
  2. Linear justification starts from epistemically sound premises (*foundationalism*), or,
  3. Linear justification has a circular structure (*circularity*).
- A system with **infinite regress** is not pragmatic; on the other hand, circular reasoning **stagnates the epistemic status** of premises. This leaves out **foundationalism** as a suitable justificatory structure for knowledge.
- Foundationalism is a sort of **logical atomism** where the epistemic atoms are *foundational premises* and more complex premises *emerge* at the global level.
- Foundationalist approaches to empiricism, rationalism and idealism run into common issues, such as the **problem of justification of foundational claims**.

The givenist is caught in a fundamental dilemma: if [their] intutions or immediate apprehensions are construed as cognitive, then they will be both capable of giving justification and in need of it themselves; if they are non-cognitive, then they do not need justification but are also apparently incapable of providing it. This, at bottom, is why epistemological givenness is a myth. — Laurence BonJour

- A solution to the problems of foundationalism is to develop **nonlinear justification**. In this picture, premises **co-justify** each other at the *local level*, such that the *global* structure is 'coherent'.
- **Globally circular chains** of justification are now not a problem, as epistemic stagnation is an artefact of linear justification.
- At a local level of knowledge, one can decide a direction for justification (such as via psychological factors) such that there are foundations and no circles of justification.
- At the global level of knowledge, there need not be strict foundations, and premises can be coherently explained by non-linear co-justification from other premises, in a manner over and above that of foundationalism.
- Thus, foundations are different **local modes** of a coherentist system of knowledge! This is consistent with Gödel's incompleteness theorems for sufficiently complicated formal systems.

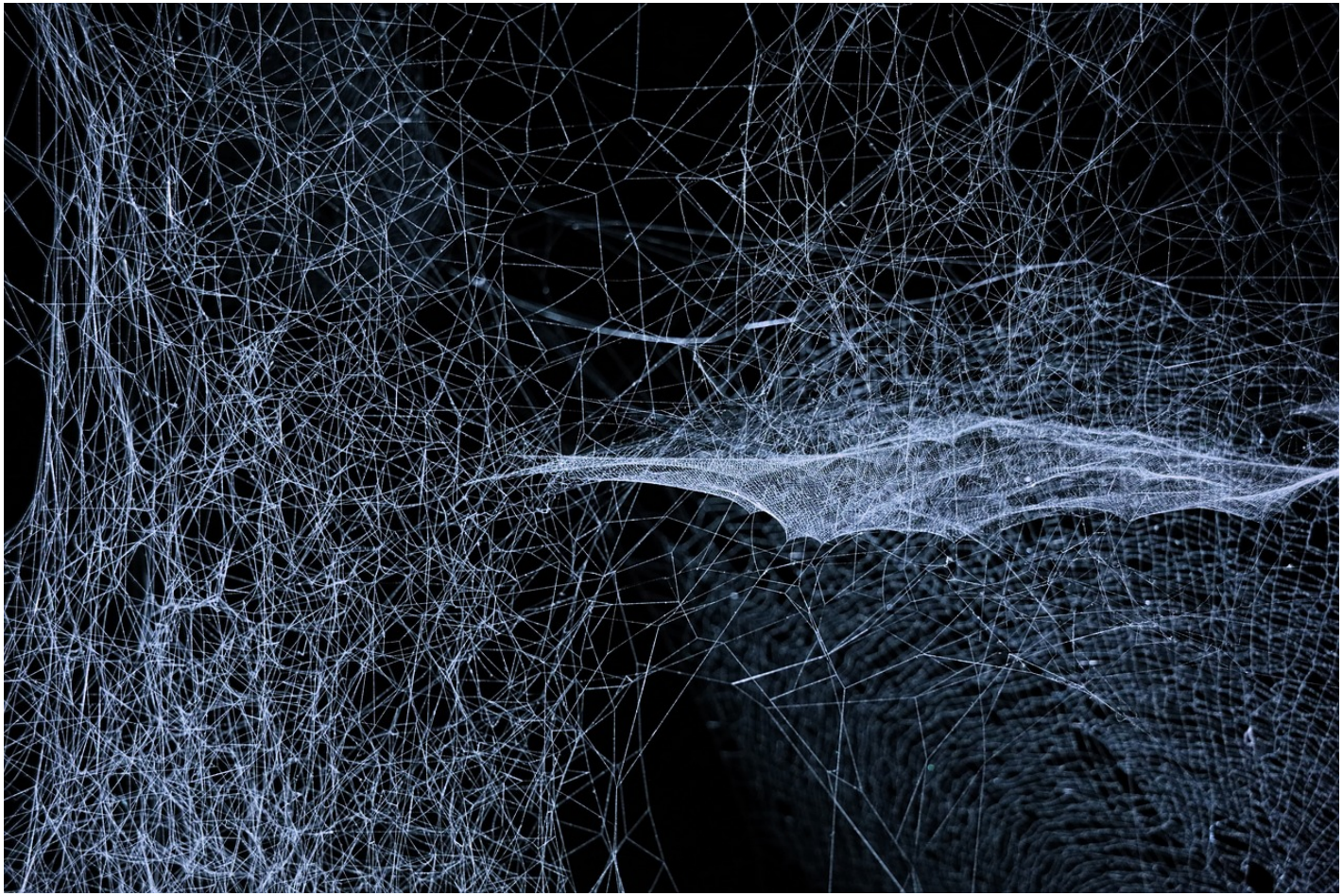


Figure: A coherentist **model** of knowledge would be an *undirected graph* (with the *nodes* being premises and *edges* being nonlinear justification) such that there exist multiple subsets which are *trees* (i.e. contain no cycles). These trees generate **local justificatory flow**. The *foundations* of the model are then characterized by these subtrees. The system is coherent if and only if its foundational subsets are internally consistent; however, they need not be complete.

Ontological positions in science

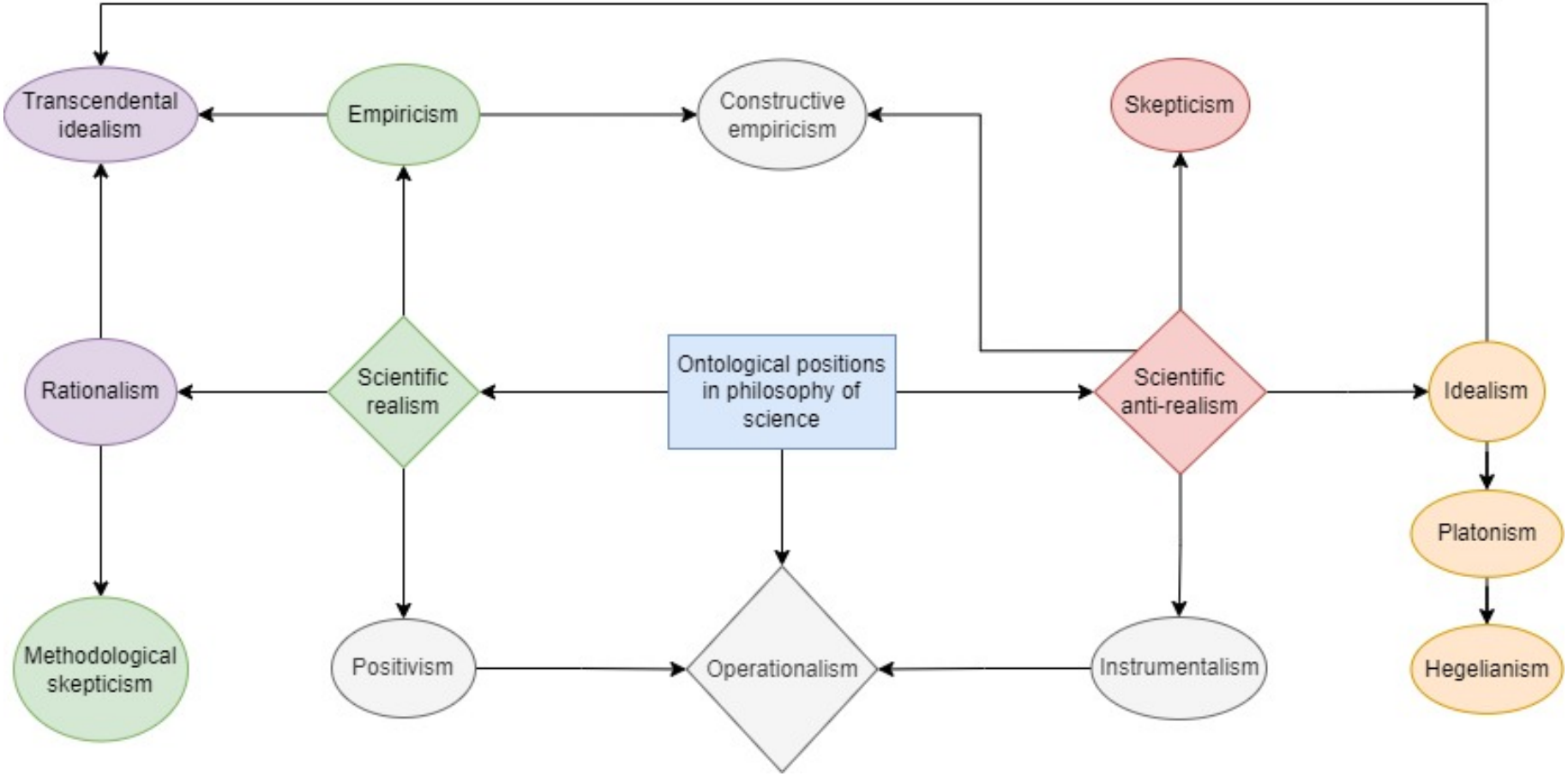


Figure: A rough picture of different positions regarding the ontological status of science. Notice that the categories of ideas involved are not mutually exclusive.

Scientific realism

- Scientific realism is a **positive epistemic attitute** towards the contents of scientific theories.
- A traditional scientifict realist typically **commits to the truth** of various comabinations of the following metaphysical claims:
  1. **mind-independent material objects** represented by abstractions in scientific models (*objectivity*)
  2. **causal processes** describing the coming-to-be of one physical object from another (*causation*)
  3. **mathematical laws** which effectively govern the above processes but structurally live in the world of concepts/forms (*perennialism*)
  4. the ability to take rational statements about the above kinds of objects at **literal face value** (*positivism*)
- There are multiple versions and degrees of scientific realism, and the above commitments are not sufficient or necessary to characterize said versions of realism. Some critical realists, for example, can disagree with (3), (4) and to a degree, (1), as we shall see.
- One way of looking at the above claims is to organize them as different aspects of scientific realism:
  1. **Metaphysical**: objectivity, causation, perennialism
  2. **Semantical**: positivism
  3. **Epistemological**: the implicit view that the study of the objects of science is faithful knowledge

Epistemic gradation in scientific realism

- Usually, accepting all three of the above aspects of scientific realism is its strongest version — **rationalism**. On the other hand, dropping the epistemological commitment amounts to **skepticism**. Skepticism is itself graded:
  1. **Strong skepticism**: science, in no stage, can constitute incorrigible knowledge.
  2. **Weak skepticism**: science in its current stage is likely corrigible but progressive versions will either keep approaching 'the truth' or eventually reach it
- Notice that weak skepticism makes a metaphysical claim about truth, which can range from absolutism to monisitic relativism (which is at the epistemic end of the relevant ontological gradation).
- The version of realist, weak skepticism compatible with the metaphysical-semantical dimension of realism is a form of **pragmatism**:
  1. In the most ontic version of such pragmatism, the semantics of non-empirical entities constitutes useful knowledge in *effective theories* but not *fundamental theories* (**effectivism**).
  2. On the epistemic end of realist pragmatism, one may drop positivism. This leads to **instrumentalism**, where the semantics of non-empirical statements is not truth-evaluable.

- Thus, under an *ontic* $\rightarrow$ *epistemic* gradation, realism transitions to anti-realism:

$$
\begin{tikzcd}
\textit{Magical realism} & \textit{Pragmatism} & \textit{Humeanism} \\
\text{Rationalism} \arrow[u, dashed] \arrow[r] & \text{Weak skepticism} \arrow[u, dashed] \arrow[r] \arrow[d] & \text{Strong skepticism} \\
\textit{Pragmatism} & \text{Absolutism} \arrow[l, dashed] \arrow[d] & \textit{Post-structuralism} \arrow[u, bend right] \\
\textit{Structuralism} & \text{Monistic relativism} \arrow[l, dashed] \arrow[r] & \text{Pluralistic relativism} \arrow[u, dashed]
\end{tikzcd}
$$

Figure: An epistemic gradation of scientific realism. Dotted arrows represent theoretical justification rather than epistemic gradation.

Scientific anti-realism

- In scientific realist views of the ontology of science, there is typically a notion of ontological objects which physically exist. These are called the **beables** of the theory being interpreted as realist (this term was invented by Irish mathematical physicist John Stewart Bell, 1928-1990).
- For example, in physicalist realism, the beables are mind-independent objects undergoing causal flux.
- In scientific anti-realism, such material objects are usually *not* beables. For example, in **subjective idealism**, the beables are:
  1. a **transcendental observer**
  2. abstract **concepts** accessible to the observer

  The above are often collectively called the **mind**. Mind-independent beables are replaced by the epistemically weaker notion of **observables**, which are **mental events** such as measurements of so-called external events. Even if observed events are somewhat external to the mind, these events **come into being** by virtue of **perception** of the mind.
- Examples of subjective idealists are Plato (427-348 BC), Plotinus (204/5-270), Augustine (354-430), George Berkeley (1685-1753) in his thesis "**to be is to be perceived**", J. G. Fichte (1762–1814) and G. W. F. Hegel (1770-1831).

Epistemic gradation in scientific anti-realism

- On the strongest end of anti-realism[1], there *are* no beables to begin with, i.e., there is no well-defined observer or abstract concept. This is **hard skepticism**, a proponent of which was David Hume (1711-1776).
- On the weakest end of anti-realism, **physical reality is a beable, but not cognizable as things-in-themselves**. Instead, physical contents of reality can only be **phenomenologically** organized given *a priori* mental/cognitive architecture. This is **transcendental idealism**, realized in the philosophies of Immanuel Kant (1724-1804) and Arthur Schopenhauer[2] (1788-1860).
- A stronger variant of transcendental idealism is **objective idealism**: Being is physical but metaphysically isomorphic to Thought. Part of this is **tychism**, the idea that "*matter is effete mind, inveterate habits becoming physical laws*". Thus, chance is an objective feature of the universe, resulting in **nomological novelty**. This philosophy was developed by Charles Sanders Peirce (1839-1914), who also founded pragmatism.

[1] For brevity, absolute idealism and speculative idealism have been omitted.

[2] Schopenhauer did think of the thing-in-itself as cognizable via the Will, which is somewhat reciprocal to perception: the mind becomes the pure subject of knowing the world aethetically.

- Without the tychism part (and instead the much stricter *Principle of Sufficient Reason*), G. W. Leibniz's (1646-1716) **monadology** is a form of objective idealism.
- With regard to the mind-body problem, Baruch Spinoza (1632-1677) developed a form of objective idealism called **neutral monism**, in which there is only one infinite substance — God — and mental life and physical reality are different modes of God.
- In Berkeley's theological addendum to his subjective idealist thesis, there is an objective reality to objects when perceived by God, elevating his metaphysics to objective idealism.
- F. W. J. Schelling (1775-1854) was an objective idealist who took transcendental idealism a step further in his thesis that the mental cognizes the physical to the extent that their distinction is purely psychological.

$$
\begin{tikzcd}
 & \textit{Kantianism} & \textit{Platonism} \arrow[r, dashed] & \textit{Hegelianism} & \textit{Humeanism} \\
 & \text{Transcendental idealism} \arrow[u, dashed] \arrow[r] \arrow[d] & \text{Subjective idealism} \arrow[u, dashed] \arrow[r] & \text{Skepticism} \arrow[u, dashed] & \\
 & \text{Objective idealism} \arrow[ur] \arrow[r, dashed] & \textit{Monadology} & &
\end{tikzcd}
$$

Figure: A landscape of scientific anti-realism.

Dualism

- In scientific realism, physical objects gain ontological precedence over mental events, while in anti-realism, it is vice-versa.
- In dualism, **both physical and mental reality are comparably ontological**. Equivalently, dualism is itself a graded family of philosophies, which in the strongest form assumes Cartesian **substance dualism** and in the weakest end, adopts one or more of the following:
  1. **Predicate dualism**: there are only physical phenomena, but mental-talk is not reducible to physical-talk.
  2. **Idealistic dualism**: there are physical and mental phenomena, and mental-talk cognizes physical-talk. e.g. transcendental idealism.
  3. **Physicalistic dualism**: there are physical and mental events, and mental-talk is roughly translatable to physical-talk. e.g. post-analytic positivism.
- Mentalistic and physicalistic dualism are forms of **double knowledge theories**, wherein physicalism and phenomenalism are inter-translatable via a **bridge theory**.
- In the stronger form of predicate dualism, mental life can play a non-causal role that cannot be reduced to physical-talk (**property dualism**). This is consistent with physicalism but not reductionism and is realized in **epiphenomenalism**.

- Substance dualism is refutable on grounds of the **interaction problem**. Moreover, mental events being accessed as knowledge by acquintance, as opposed to physical events being accessed as knowledge by description[3] (**knowledge gap**), does not necessarily rule out double knowledge (U. T. Place, 1924-2000). To commit to such an argument would be a **category mistake** (Gilbert Ryle, 1900-1976).
- However, predicate and property dualism are consistent with physicalism, since mental and physical events enjoy their own causal closures, without causal contamination. However, there could not only be mental-talk that is non-reducible to physical terms, but there could also be mental properties such as **qualia** (the objects of phenomenological experience) which supervene on physical objects.

$$\begin{tikzcd}
\text{Events} \arrow[r, "M \text{ properties}"] \arrow[d, "P \text{ properties}"'] & M \text{ talk} \\
P \text{ talk} \arrow[ur, dashed, no head] &
\end{tikzcd}$$

Figure: The problem of dualism in an event ontology setting.

[3] These epistemic terms were coined by **Bertrand Russell** (1872-1970).

Event ontology

- **W. V. O. Quine's** (1908-2000) **dictum of ontological commitment**: *stating a sentence is committing to the ontological existence of its quantificationally bound variables*.
- In **Plato's theory of forms**, stating a sentence is committing to the existence of quantified objects, as well as their predicates/forms. Thus, traditional Platonic ontology restricts itself to **second-order logic**.
  For example, in Platonic ontology, one would translate 'there is a red (R) truck (x) such that it travelled (T) on an object(y) called a path (P)' as,

  $$\exists X, Y : \exists x \in X \exists R \in {\mathbb{Z}_2}^X \exists y \in Y \exists P \in {\mathbb{Z}_2}^Y \exists T \in {\mathbb{Z}_2}^{X \times Y} : R(x) \wedge P(y) \wedge T(x, y)$$

  where X could be the vehicle-form and Y, the functional form of paths. In Platonic semantics, we commit to the existence of all the existential entities above.
- **Russell's theory of types** extends the above idea to **higher-order logic** and therefore, commitment to the existence of higher-order sentential objects.
- In **Donald Davidson's** (1917-2003) **event ontology**, *events are property instantiations* (Jaegwon Kim, 1934-2019) to be quantified over, therefore, events are ontological objects in **first-order sentences**.

- Davidson's event ontology leads to the novel view that **only physical events are beables**, not material objects. For example, the sentence under Plato's theory of forms would now be translated as,

$$\exists x \in X \, \exists y \in Y : R(x) \wedge P(y) \wedge T(x, y)$$

- An interpretation of material objects, therefore, is that they are idealizations of events approximately satisfying object-like properties. One would still quantify over events in this picture, therefore, **what 'exists' is not material objects but collections of roughly well-behaved events**. Therefore, this is a form of monistic atomism, where the atoms are events and they have the same causal structure.
- Such event ontology is compatible with **flux**, which is the opposite of Parmenides' (late sixth/early fifth century BC) eternalism. Here, not only are events atoms of ontology, but they are also units of change, which is the true ontological nature of the world.
- Therefore, **there is no material being, but only event-like becoming**. Such philosophy was conceived of by Heraclitus (active in 500 BC), G. W. Leibniz and Martin Heidegger (1889-1976); Alfred North Whitehead (1861-1947) offers a mature version of flux-based ontology, called **process philosophy**.

Indexicality

- Sometimes, one can interpret propositional attitude statements as **indexed by quantified agents**. For example, if $\mathcal{S}$ is a set of subsystems in a causal system (including agents) and we are describing the example statement previously described in Davidson's event ontology,

$$\exists S, T, \cdots \in \mathcal{S}\, \exists x \in X\, \exists y \in Y : K_{S,T,\ldots}\left(R(x) \wedge P(y) \wedge T(x,y)\right)$$

  The truth-evaluability of indexical statements depends on the context, viz. the agents in question.
- In idealism, the non-agent events being quantified over are interpreted as existing in virtue of the agents.
- On the other hand, in **materialistic relativism**, one interprets the agents as further collections of quantified events, with agent-like properties. This is a form of critical realism, and a foreshadow of relationalism in quantum mechanics.
- The sentential structure of events also modifies dualism. Instead of mental/physical predicate-talk, whose existence we need not commit to anymore, **the question becomes whether there are two distinct classes of events: observables and agents**.

Causal metaphysics

- Two important theses in Hume's anti-realist causal metaphysics are:
  1. Inferences to do with material objects are **inductive**, i.e., contingent (not necessary in all *logical worlds*).
  2. Such inferences are typically examples of **causal inference**, which is itself contingent.

  Therefore, causation is not logical in the sense of deductions performed on universal concepts.
- Rather, when events of type A are **necessarily** observed in **conjunction** with antecedent events of type B in the past, these *observations* **mentally cause** an observer to form the *idea* of B whenever A is observed in the future. In the hybrid language of epistemic and linear temporal modal logics:

$$\begin{aligned}
\psi(A,B) &\iff (A \preceq B) \wedge (A \implies B) \\
\varphi(\tau) &\iff (t = \tau) \\
\forall \tau : B_S\left(\psi(A,B)\, \mathcal{W}\, \varphi(\tau)\right) &\rightsquigarrow \bigcirc B_S(\Box\psi(A,B)) \\
B_S(\Box\psi(A,B)) &\rightsquigarrow \bigcirc B_S(\Box\psi(A,B))
\end{aligned}$$

- In the above causal process, LHS and RHS involve belief states in S and the causal mechanism connecting them is an *intentional* substitution in S. The contingence of the contents of these beliefs is captured in the first transition of the *weak until* relation ($\mathcal{W}$) to the *global* condition ($\Box$).

Sentence	Operator	Tree diagram
$\bigcirc\psi$	*Next*	$\bullet \rightarrow \psi \rightarrow \bullet \dashrightarrow$
$\Diamond\psi$	*Finally*	$\bullet \dashrightarrow \bullet \rightarrow \psi \rightarrow \bullet \dashrightarrow$
$\Box\psi$	*Globally*	$\psi \dashrightarrow$
$\psi\,\mathcal{U}\,\varphi$	*Until*	$\psi \dashrightarrow \psi \rightarrow \varphi \rightarrow \bullet \dashrightarrow$
$\psi\,\mathcal{W}\,\varphi$	*Weak until*	$\psi \dashrightarrow \psi \rightarrow \varphi \rightarrow \bullet \dashrightarrow$ $\psi \dashrightarrow$
$\psi\,\mathcal{R}\,\varphi$	*Release*	$\varphi \dashrightarrow \varphi \dashrightarrow \varphi, \psi \rightarrow \bullet \dashrightarrow$ $\varphi \dashrightarrow$
$\psi\,\mathcal{M}\,\varphi$	*Strong release*	$\varphi \dashrightarrow \varphi \dashrightarrow \varphi, \psi \rightarrow \bullet \dashrightarrow$

Figure: Modal operators in Linear Temporal Logic (LTL).

Anscombian causality

- According to **G. E. M. Anscombe** (1919-2001), Hume's notion of causation is too rigid. Instead of a necessary type contiguity for causation, Anscombe's description is a metaphor for a **retrospective probabilistic space-token model** for causality:

$$\widehat{\mathsf{cor}}\,(a,b) \iff \underbrace{\mathsf{pr}\,(a|b) - \mathsf{pr}\,(\overline{a}|b)}_{\mathsf{cor}(a,b)} \in (0,1]$$

$$\psi(a,b) \iff (a \preceq b) \wedge \widehat{\mathsf{cor}}\,(a,b) \wedge \widehat{\mathsf{cor}}\,\left(\overline{a},\overline{b}\right)$$

$$a \rightsquigarrow b \iff \Diamond\psi(a,b)$$

This *extensionally* leads to the epistemic state updates:

$$B_S(\Diamond\psi(a,b)) \rightsquigarrow \bigcirc B_S(a \to b)$$

$$B_S(a \to b) \rightsquigarrow \bigcirc B_S(a \to b)$$

- In other words, two partially ordered token events a, b are causally connected when:
  1. The consequent b is **correlated** to the antecedent a.
  2. If the consequent is antecedented by any intermediate event c, the intermediate event c is itself antecedented by a.

Causal sets

- In the language of partially ordered sets, if $a, b, c, \cdots \in X$, then $(X, \preceq)$ is a **poset**, i.e. satisfies the following:
  1. **Reflexivity**: $\forall a \in X : a \preceq a$.
  2. **Antisymmetry**: $\forall a, b \in X : (a \preceq b) \wedge (a \succeq b) \implies (a = b)$.
  3. **Transitivity**: $\forall a, b, c \in X : (a \preceq b) \wedge (b \preceq c) \implies (a \preceq c)$
- Let (X, Σ, pr) be a probability measure. Then, we have:

$$\begin{tikzcd}
\mathsf{im}_{\mathsf{pr}}\left(\Sigma/\ker\left(\mathsf{pr}\right)\right) \times \mathsf{im}_{\mathsf{pr}}\left(\Sigma/\ker\left(\mathsf{pr}\right)\right) \arrow[r, "\leq"] & \mathbb{Z}_2 \\
\Sigma/\ker\left(\mathsf{pr}\right) \times \Sigma/\ker\left(\mathsf{pr}\right) \arrow[u, "\mathsf{im}_{\mathsf{pr}} \otimes \mathsf{im}_{\mathsf{pr}}"] \arrow[ur, "\preceq_{\mathsf{pr}}"'] &
\end{tikzcd}$$

(a) In the category of sets (Set)

$$\begin{tikzcd}
\left(\mathsf{pr}\left(\Sigma/\ker\left(\mathsf{pr}\right)\right), \leq\right) \arrow[r, bend left, "\mathsf{preim}_{\mathsf{pr}}"] & \left(\Sigma/\ker\left(\mathsf{pr}\right), \preceq_{\mathsf{pr}}\right) \arrow[l, bend left, "\mathsf{im}_{\mathsf{pr}}"]
\end{tikzcd}$$

(b) In the category of posets (Ord)

- Consequently, $(\Sigma/\ker(\mathsf{pr}), \preceq_{\mathsf{pr}})$ in Pos (i.e., is a poset) where $\preceq_{\mathsf{pr}} = \leq \circ (\mathsf{im}_{\mathsf{pr}} \otimes \mathsf{im}_{\mathsf{pr}})$,
  1. *Reflexivity*:

$$\begin{aligned} \forall A \in \Sigma/\ker(\mathsf{pr}) : \mathsf{pr}(A) &\leq \mathsf{pr}(A) \\ &\implies A \preceq_{\mathsf{pr}} A \end{aligned}$$

  2. *Antisymmetry*:

$$\begin{aligned} \forall A, B \in \Sigma/\ker(\mathsf{pr}) : (A \preceq_{\mathsf{pr}} B) &\wedge (B \preceq_{\mathsf{pr}} A) \\ &\implies (\mathsf{pr}(A) \leq \mathsf{pr}(B)) \wedge (\mathsf{pr}(B) \leq \mathsf{pr}(A)) \\ &\implies \mathsf{pr}(A) = \mathsf{pr}(B) \\ &\implies A \sim B \end{aligned}$$

  3. *Transitivity*:

$$\begin{aligned} \forall A, B, C \in \Sigma/\ker(\mathsf{pr}) : (A \preceq_{\mathsf{pr}} B) &\wedge (B \preceq_{\mathsf{pr}} C) \\ &\implies (\mathsf{pr}(A) \leq \mathsf{pr}(B)) \wedge (\mathsf{pr}(B) \leq \mathsf{pr}(C)) \\ &\implies \mathsf{pr}(A) \leq \mathsf{pr}(C) \\ &\implies A \preceq_{\mathsf{pr}} C \end{aligned}$$

Conditional probability

- **Conditional probability** is one of the Σ^*-indexed probability measures,

$$\Sigma^* = \Sigma \setminus \ker(\mathsf{pr})$$
$$\mathsf{pr}(\cdot|\cdot) : \Sigma \times \Sigma^* \to [0,1]$$

$$\boxed{\forall A \in \Sigma\, \forall B \in \Sigma^* : \mathsf{pr}(A|B) = \frac{\mathsf{pr}(A \cap B)}{\mathsf{pr}(B)}}$$

- $\forall B \in \Sigma^* : \left(\Sigma/\ker(\mathsf{pr}(\cdot|B)), \preceq_{\mathsf{pr}(\cdot|B)}\right)$ in Pos, where $\preceq_{\mathsf{pr}(\cdot|B)} = \leq \circ \left(\mathsf{im}_{\mathsf{pr}(\cdot|B)} \otimes \mathsf{im}_{\mathsf{pr}(\cdot|B)}\right)$.
- A useful identity is **Bayes' theorem**:

$$\forall A, B \in \Sigma^* : \mathsf{pr}(A|B)\,\mathsf{pr}(B) = \mathsf{pr}(B|A)\,\mathsf{pr}(A) \qquad [= \mathsf{pr}(A \cap B)]$$

Correlation

- As defined earlier, **correlation** (cor) is the Σ-odd part of conditional probability (and hence with range $[-1, 1]$) and $\widehat{\text{cor}}$ is its signature. As it turns out, $\text{cor}(\cdot, B)$ does *not* generate signed measures on $\Sigma/\ker(\text{cor}(\cdot, B))$ for any $B \in \Sigma^*$,

$$\forall A_1, A_2 \in \Sigma/\ker(\text{cor}(\cdot, B))\ \forall B \in \Sigma^* : A_1 \cap A_2 = \emptyset$$

$$\begin{aligned}
\implies \text{cor}(A_1 \cup A_2, B) &= \text{pr}(A_1 \cup A_2 | B) - \text{pr}\left(\overline{A_1 \cup A_2} \middle| B\right) \\
&= \frac{\text{pr}((A_1 \cup A_2) \cap B)}{\text{pr}(B)} - \frac{\text{pr}\left(\overline{A}_1 \cap \overline{A}_2 \cap B\right)}{\text{pr}(B)} \\
&= \frac{\text{pr}((A_1 \cap B) \cup (A_2 \cap B))}{\text{pr}(B)} - \frac{\text{pr}\left((\overline{A}_1 \cap B) \cap (\overline{A}_2 \cap B)\right)}{\text{pr}(B)} \\
&= \frac{\text{pr}(A_1 \cap B) + \text{pr}(A_2 \cap B)}{\text{pr}(B)} \\
&= \frac{\text{pr}\left((\overline{A}_1 \cap B) \cup (\overline{A}_2 \cap B)\right) - \text{pr}\left(\overline{A}_1 \cap B\right) - \text{pr}\left(\overline{A}_2 \cap B\right)}{\text{pr}(B)}
\end{aligned}$$

$$
\begin{aligned}
&= \mathsf{pr}\left(A_1|B\right) + \mathsf{pr}\left(A_2|B\right) - \mathsf{pr}\left(\overline{A}_1\middle|B\right) - \mathsf{pr}\left(\overline{A}_2\middle|B\right) + \frac{\mathsf{pr}\left(\left(\overline{A}_1 \cup \overline{A}_2\right) \cap B\right)}{\mathsf{pr}\left(B\right)} \\
&= \mathsf{cor}\left(A_1, B\right) + \mathsf{cor}\left(A_2, B\right) + \frac{\mathsf{pr}\left(\left(\overline{A_1 \cap A_2}\right) \cap B\right)}{\mathsf{pr}\left(B\right)} \\
&= \mathsf{cor}\left(A_1, B\right) + \mathsf{cor}\left(A_2, B\right) + \frac{\mathsf{pr}\left(\overline{\emptyset} \cap B\right)}{\mathsf{pr}\left(B\right)} \\
&= \mathsf{cor}\left(A_1, B\right) + \mathsf{cor}\left(A_2, B\right) + 1
\end{aligned}
$$

- A useful property of cor is **extended Bayes' theorem**:

$$
\begin{aligned}
\mathsf{cor}\left(A, B\right)\mathsf{pr}\left(B\right) &= \mathsf{pr}\left(B|A\right)\mathsf{pr}\left(A\right) - \mathsf{pr}\left(\overline{A} \cap B\right) \\
&= \mathsf{pr}\left(B|A\right)\mathsf{pr}\left(A\right) + \mathsf{pr}\left(A \cup \overline{B}\right) - 1 \\
&= \mathsf{pr}\left(B|A\right)\mathsf{pr}\left(A\right) + \mathsf{pr}\left(A\right) + \mathsf{pr}\left(\overline{B}\right) - \mathsf{pr}\left(A \cap \overline{B}\right) - 1 \\
&= \mathsf{pr}\left(B|A\right)\mathsf{pr}\left(A\right) + \mathsf{pr}\left(A\right) - \mathsf{pr}\left(B\right) - \mathsf{pr}\left(\overline{B}\middle|A\right)\mathsf{pr}\left(A\right) \\
&= \mathsf{cor}\left(B, A\right)\mathsf{pr}\left(A\right) + \mathsf{pr}\left(A\right) - \mathsf{pr}\left(B\right)
\end{aligned}
$$

$$
\forall A, B \in \Sigma^* : \left[\mathsf{cor}\left(A, B\right) + 1\right]\mathsf{pr}\left(B\right) = \left[\mathsf{cor}\left(B, A\right) + 1\right]\mathsf{pr}\left(A\right)
$$

Relative correlation

- It was seen previously that cor is *almost* subadditive, in the same sense affine maps are *almost* linear maps. To fix this, one may define a Σ^*-indexed family of **relative correlation** functions:

$$\boxed{\forall C \in \Sigma^* : \mathsf{cor}_C : \begin{cases} \Sigma \times \Sigma^* & \to [-1,1] \\ (A,B) & \mapsto \frac{1}{2}\mathsf{cor}\,(A,B) - \frac{1}{2}\mathsf{cor}\,(A,C) \end{cases}}$$

- Indeed, $\forall B, C \in \Sigma^* : (X, \Sigma, \mathsf{cor}_C\,(\cdot, B))$ in $\mathsf{M}_{\mathsf{pr,sgn}}$, i.e. the category of signed probability spaces,
  1. $\forall A \in \Sigma\, \forall B, C \in \Sigma^* : \mathsf{cor}_C\,(A,B) \in [-1,1]$.
  2. $\forall B, C \in \Sigma^* : \mathsf{cor}_C\,(\emptyset, B) = \mathsf{cor}\,(\emptyset, B) - \mathsf{cor}\,(\emptyset, B) = 0$.
  3. Suppose $A_1, A_2 \in \Sigma$ with $A_1 \cap A_2 = \emptyset$. Then, subadditivity is obtained as follows:

$$\forall B, C \in \Sigma^* : \mathsf{cor}_C\,(A_1 \cup A_2, B) = \frac{1}{2}\mathsf{cor}\,(A_1 \cup A_2, B) - \frac{1}{2}\mathsf{cor}\,(A_1 \cup A_2, C)$$
$$= \frac{1}{2}\mathsf{cor}\,(A_1, B) + \frac{1}{2}\mathsf{cor}\,(A_2, B) - \cancel{1} - \frac{1}{2}\mathsf{cor}\,(A_1, C) - \frac{1}{2}\mathsf{cor}\,(A_2, C) + \cancel{1}$$
$$= \mathsf{cor}_C\,(A_1, B) + \mathsf{cor}_C\,(A_2, B)$$

Principal correlation

- When $C = X$, we get **principal (relative) correlation**,

$$\boxed{\forall A \in \Sigma\, \forall B, C \in \Sigma^* : \mathsf{cor}_X\left(A, B\right) = \frac{1}{2}\mathsf{cor}\left(A, B\right) - \mathsf{pr}\left(A\right) + \frac{1}{2}}$$

  It follows, $\forall B \in \Sigma^* : \left(X, \Sigma, \mathsf{cor}_X\left(\cdot, B\right)\right)$ in $\mathsf{M}_{\mathsf{pr,sgn}}$ is induced by the M_{pr} object $\left(X, \Sigma, \mathsf{pr}\right)$.
- Once again[4], $\forall B \in \Sigma^* : \left(\Sigma / \ker\left(\mathsf{cor}_X\left(\cdot, B\right)\right), \preceq_{\mathsf{cor}_X(\cdot, B)}\right)$ in Pos. Now, $\widehat{\mathsf{cor}}_X$ does not itself generate a poset[5], as,

$$\begin{aligned}
\forall A \in \Sigma^* : \mathsf{cor}_X\left(A, A\right) &= \frac{1}{2}\mathsf{cor}\left(A, A\right) - \mathsf{pr}\left(A\right) + \frac{1}{2} \\
&= \frac{1}{2} - \mathsf{pr}\left(A\right) + \frac{1}{2} \\
&= 1 - \mathsf{pr}\left(A\right) \in [0, 1) \neq (0, 1] \\
&\not\Longrightarrow \widehat{\mathsf{cor}}_X\left(A, A\right)
\end{aligned}$$

[4] we familiarly define $\preceq_{\mathsf{cor}_X(\cdot, B)} = \leq \circ \left(\mathsf{im}_{\mathsf{cor}_X(\cdot, B)} \otimes \mathsf{im}_{\mathsf{cor}_X(\cdot, B)}\right)$.
[5] where $\widehat{\mathsf{cor}}_X\left(A, B\right) \iff \mathsf{cor}_X\left(A, B\right) \in (0, 1]$.

Principal conjugate correlation

- A tempting attempt to resolve this issue is to consider a new quantity, the **principal conjugate correlation**,

$$\boxed{\forall A \in \Sigma\, \forall B \in \Sigma^* : \overline{\mathsf{cor}}_X\left(A, B\right) = 1 - \mathsf{cor}_X\left(A, B\right) = \mathsf{pr}\left(A\right) - \frac{1}{2}\mathsf{cor}\left(A, B\right) + \frac{1}{2}}$$

Equivalently, defining **conjugate correlation** as $\overline{\mathsf{cor}}\left(A, B\right) = 1 - \mathsf{cor}\left(A, B\right)$,

$$\boxed{\forall A \in \Sigma\, \forall B \in \Sigma^* : \overline{\mathsf{cor}}_X\left(A, B\right) = \mathsf{pr}\left(A\right) + \frac{1}{2}\overline{\mathsf{cor}}\left(A, B\right)}$$

- Thus, we obtain the **conjugate version of the extended Bayes' theorem**:

$$\forall A, B \in \Sigma^* : \left[\overline{\mathsf{cor}}\left(A, B\right) - 2\right] \mathsf{pr}\left(B\right) = \left[\overline{\mathsf{cor}}\left(B, A\right) - 2\right] \mathsf{pr}\left(A\right)$$

- Similarly, one finds the **principal conjugate version of the extended Bayes' theorem**:

$$\begin{aligned}\overline{\mathsf{cor}}_X(A,B)\,\mathsf{pr}(B) &= \mathsf{pr}(A)\,\mathsf{pr}(B) + \frac{1}{2}\overline{\mathsf{cor}}(A,B)\,\mathsf{pr}(B)\\ &= \mathsf{pr}(A)\,\mathsf{pr}(B) + \frac{1}{2}\left[\frac{(\overline{\mathsf{cor}}(B,A)-2)\,\mathsf{pr}(A)}{\mathsf{pr}(B)} + 2\right]\mathsf{pr}(B)\\ &= \mathsf{pr}(B)\,\mathsf{pr}(A) + \frac{1}{2}\overline{\mathsf{cor}}(A,B)\,\mathsf{pr}(A) - \mathsf{pr}(A) + \mathsf{pr}(B)\\ &= \overline{\mathsf{cor}}_X(B,A)\,\mathsf{pr}(A) - \mathsf{pr}(A) + \mathsf{pr}(B)\end{aligned}$$

$$\forall A,B \in \Sigma^* : [\overline{\mathsf{cor}}_X(A,B) - 1]\,\mathsf{pr}(B) = [\overline{\mathsf{cor}}_X(B,A) - 1]\,\mathsf{pr}(A)$$

- This leads to the elegant **principal version of the extended Bayes' theorem**:

$$\forall A,B \in \Sigma^* : \mathsf{cor}_X(A,B)\,\mathsf{pr}(B) = \mathsf{cor}_X(B,A)\,\mathsf{pr}(A)$$

- It turns out that $\widehat{\overline{\mathsf{cor}}}_X$ induces a **tolerance** (i.e., a reflexive, antisymmetric relation) on Σ^* once we quotient it out by an appropriate equivalence relation:

$$\left(\Sigma^* / \ker(\mathsf{pr}), \widehat{\overline{\mathsf{cor}}}_X\right) \text{ in } \mathsf{Tol}$$

1. *Reflexivity*:

$$\forall A \in \Sigma^* / \ker(\mathsf{pr}) : \overline{\mathsf{cor}}_X(A, A) = \mathsf{pr}(A) \in (0, 1] \implies \widehat{\overline{\mathsf{cor}}}_X(A, A)$$

2. *Antisymmetry*:

$$\begin{aligned}
\forall A, B \in \Sigma^* / \ker(\mathsf{pr}) : \widehat{\overline{\mathsf{cor}}}_X(A, B) \wedge \widehat{\overline{\mathsf{cor}}}_X(B, A) \\
\implies \overline{\mathsf{cor}}_X(A, B) - 1 \in [0, 1), \text{ and,} \\
\overline{\mathsf{cor}}_X(B, A) - 1 \in [0, 1) \\
\implies \frac{(\overline{\mathsf{cor}}_X(B, A) - 1)\,\mathsf{pr}(A)}{\mathsf{pr}(B)} \in [0, 1), \text{ and,} \\
\frac{(\overline{\mathsf{cor}}_X(A, B) - 1)\,\mathsf{pr}(B)}{\mathsf{pr}(A)} \in [0, 1) \\
\implies (\mathsf{pr}(A) \leq \mathsf{pr}(B)) \wedge (\mathsf{pr}(B) \leq \mathsf{pr}(A)) \\
\implies \mathsf{pr}(A) = \mathsf{pr}(B) \\
\implies A \sim B
\end{aligned}$$

Causation

- Correlation generates structure up to tolerance on Σ^* because it lacks the transitive nature of causal inference.
- One says A **passively causes** B when it is true to some extent that $\widehat{\text{cor}}\,(A, B)$ and $\widehat{\text{cor}}\,\left(\overline{A}, \overline{B}\right)$,

$$\boxed{\text{caus} : \begin{cases} \Sigma \times \Sigma^* \backslash X & \to \mathbb{R} \\ (A, B) & \mapsto \frac{1}{2}\text{cor}_X\,(A, B) + \frac{1}{2}\text{cor}_X\,\left(\overline{A}, \overline{B}\right) \\ & = \frac{1}{2}\text{cor}_{\overline{B}}\,(A, B) \end{cases}}$$

For $A = \emptyset$, we have,

$$\begin{aligned} \forall B \in \Sigma^* \backslash X : \text{caus}\,(\emptyset, B) &= \frac{1}{2}\text{cor}_X\,(\emptyset, B) + \frac{1}{2}\text{cor}_X\,(X, B) \\ &= \frac{1}{4} \end{aligned}$$

Therefore, the defined family of quantities cannot be measures.

Relative causation

- To solve the above problem, we define a family of **relative causation** functions in a familiar manner:

$$\boxed{\forall A \in \Sigma\, \forall B, C \in \Sigma^* \backslash X : \mathsf{caus}_C\left(A, B\right) = \frac{1}{2}\mathsf{caus}\left(A, B\right) - \frac{1}{2}\mathsf{caus}\left(A, C\right)}$$

- It turns out that $\forall B, C \in \Sigma^* \backslash X : \left(X, \Sigma, \mathsf{caus}_C\left(\cdot, B\right)\right)$ in $\mathsf{M}_{\mathsf{sgn}}$,
  1. $\forall B, C \in \Sigma^* \backslash X : \mathsf{caus}_C\left(\emptyset, B\right) = \cancel{\frac{1}{4}} - \cancel{\frac{1}{4}} = 0.$
  2. $\forall C \in \Sigma^* \backslash X : \mathsf{im}_{\mathsf{caus}_C}\left(\Sigma \times \Sigma^* \backslash X\right) = [-1, 1].$
  3. Suppose $A_1, A_2 \in \Sigma$ and $A_1 \cap A_2 = \emptyset$. Then, $\mathsf{caus}_C\left(\cdot, B\right)$ is subadditive iff $\mathsf{cor}_X\left(\cdot, B\right), \mathsf{cor}_X\left(\bar{\cdot}, \overline{B}\right)$ are subadditive, as the expression for the former involves a linear combination of terms with types from the latter. But we know that $\mathsf{cor}_X\left(\cdot, B\right)$-type quantities are measures. We need only show:

$$\begin{aligned}
&\forall B, C \in \Sigma^* \backslash X : \mathsf{cor}_X\left(\overline{A_1 \cup A_2}, B\right) \\
&= \mathsf{cor}_X\left(\overline{A}_1 \cap \overline{A}_2, B\right) \\
&= \mathsf{cor}_X\left(\overline{A}_1, B\right) + \mathsf{cor}_X\left(\overline{A}_2, B\right) - \mathsf{cor}_X\left(\overline{A}_1 \cup \overline{A}_2, B\right) \\
&= \mathsf{cor}_X\left(\overline{A}_1, B\right) + \mathsf{cor}_X\left(\overline{A}_2, B\right) - \mathsf{cor}_X\left(\cancel{\overline{A_1 \cap A_2}}, B\right) \\
&= \mathsf{cor}_X\left(\overline{A}_1, B\right) + \mathsf{cor}_X\left(\overline{A}_2, B\right) - \cancel{\mathsf{cor}_X\left(X, B\right)}
\end{aligned}$$

Interlude: σ-algebra generators and bases

- We cannot have X-relative causation, as $X \notin \Sigma^* \backslash X$. Therefore, the best we can do to fix a principal D-relative causation function, is to select the most '*causally active*' D (which is a weaker notion than the entire set X).
- To find such a D, let us set up the machinery of σ-algebra **generators**,

$$\boxed{\forall S \subseteq \Sigma : \sigma(S) = \cap \left\{ \Sigma' \subseteq \Sigma \middle| (X, \Sigma') \text{ in Meas}, S \subseteq \Sigma' \right\}}$$

By construction, σ generates σ-algebras on X due to the closure of σ-algebras under intersections,
$\forall \Sigma_1, \Sigma_2 \subseteq \Sigma : (X, \Sigma_1), (X, \Sigma_2)$ in Meas :

1. $\emptyset, X \in \Sigma_1, \Sigma_2 \implies \emptyset, X \in \Sigma_1 \cap \Sigma_2$.
2. $\forall A \in \Sigma_1 \cap \Sigma_2 : \overline{A} \in \Sigma_1, \Sigma_2 \implies \overline{A} \in \Sigma_1 \cap \Sigma_2$.
3. $\forall A, B \in \Sigma_1 \cap \Sigma_2 : A \cup B \in \Sigma_1, \Sigma_2 \implies A \cup B \in \Sigma_1 \cap \Sigma_2$.

- The **basis** $\mathcal{B}$ of Σ is the maximal element of the chain of all generators of Σ under $\supseteq$,

$$\boxed{\forall S \subseteq \Sigma : [\sigma(S) = \Sigma] \implies (S \supseteq \mathcal{B})}$$

By *Zorn's lemma*, any Σ admits a unique $\mathcal{B}$.

Principal causation

- We now set D to the $\subseteq$-maximal union of elements in $\mathcal{B}$ such that they generate a set containing X only trivially,

$$\begin{aligned} \forall S' \subset \mathcal{B} : \Gamma(S') &\iff \left(\emptyset \notin S'\right) \wedge \left(\cup S' \neq X\right) \\ \forall S' \subset \mathcal{B} : \left[\Gamma(S') &\implies \cup S' \subseteq \cup S\right] \wedge \Gamma(S) \\ D &= \cup S \end{aligned}$$

- **Principal causation** is now defined as the relation caus_D.
- Intuitively, D is 'causally maximally active' in the sense that it is the largest set in $\Sigma^* \backslash X$ such that it subsumes the most elements from $\sigma(\mathcal{B}) = \Sigma$. Therefore, on average, D is the largest arena where events in the domain of $\text{caus}_\bullet(\cdot, \cdot)$ can take place.
- It follows from the above construction that each M_{pr} object (X, Σ, pr) induces $\forall B \in \Sigma^* \backslash X : (X, \Sigma, \mathsf{caus}_D(\cdot, B))$ in $\mathsf{M}_{\mathsf{sgn}}$.

Causal structure

- We define **causal relationship** $\rightsquigarrow : \Sigma \to \Sigma^* \backslash X$ as follows. Let $(X, \preceq)$ in Pos and (X, Σ, pr) in M_{pr}. Firstly,

$$\boxed{\forall A \in \Sigma \, \forall B \in \Sigma^* \backslash X : \widehat{\mathsf{caus}}\,(A, B) \iff \mathsf{caus}\,(A, B) \in \left[\frac{1}{2}, 1\right]}$$

  Now, $A \rightsquigarrow B$ iff we have:
  1. *Partial ordering*: $A \preceq B$.
  2. *Causal transmission*: Let $U \in \Sigma, V \in \Sigma^* \backslash X$ such that $A \preceq U \preceq V \preceq B$. Then,

$$\widehat{\mathsf{caus}}\,(A, U) \wedge \widehat{\mathsf{caus}}\,(U, V) \wedge \widehat{\mathsf{caus}}\,(V, B)$$

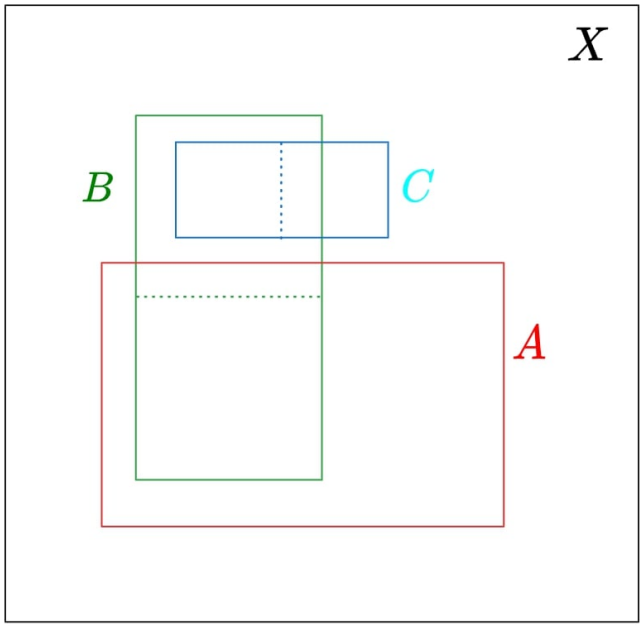
- By its very nature, $(\Sigma, \rightsquigarrow)$ in Pos,
  1. Consider any $A \in \Sigma$. Then, $A \preceq A$ and,

$$\begin{aligned}
\mathsf{caus}\,(A, A) &= \frac{1}{2}\mathsf{cor}_X\,(A, A) + \frac{1}{2}\mathsf{cor}_X\,(\overline{A}, \overline{A}) \\
&= \frac{1}{2}\,(1 - \cancel{\mathsf{pr}\,(A)}) + \cancel{\frac{1}{2}\mathsf{pr}\,(A)} \\
&\in \left[\frac{1}{2}, 1\right]
\end{aligned}$$

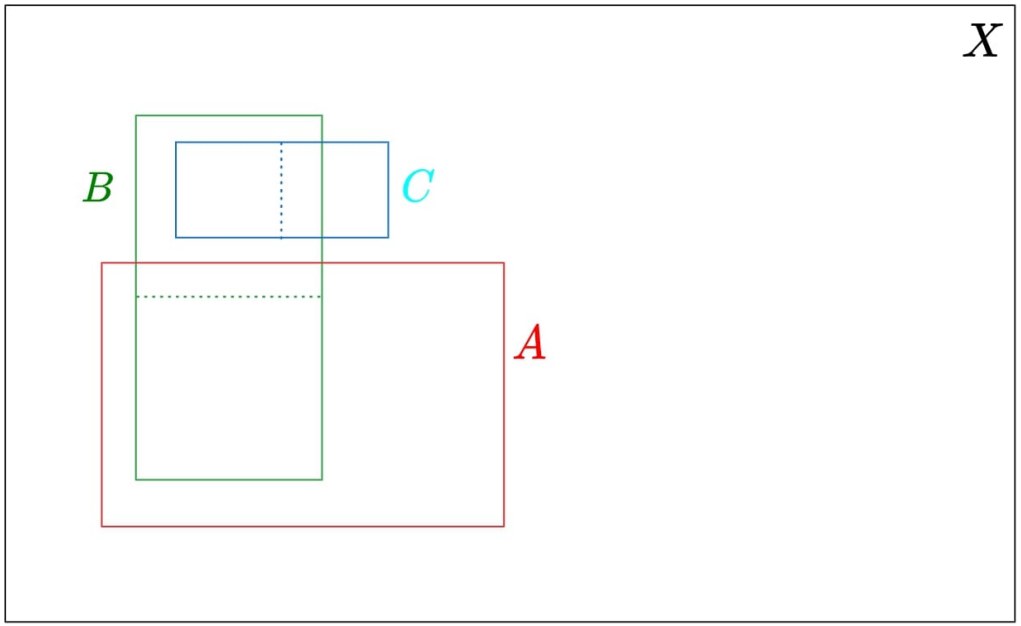
2. Suppose $A, B \in \Sigma^* \backslash X$ and $(A \rightsquigarrow B) \wedge (B \rightsquigarrow A)$. Then, by definition, $(A \preceq B) \wedge (B \preceq A) \implies A = B$.
3. Let $A \in \Sigma, B, C \in \Sigma^* \backslash X$ such that $A \rightsquigarrow B \wedge B \rightsquigarrow C$. By definition, $(A \preceq B) \wedge (B \preceq C) \implies (A \preceq C)$. Now, consider the transmission

$$\widehat{\text{caus}}\,(A, B) \wedge \widehat{\text{caus}}\,(B, C)$$

Hence, $A \rightsquigarrow C$.



(a) Non-causal correlation



(b) Causal interaction $A \rightsquigarrow B \rightsquigarrow C$

Anscombe returns

- The above causal structure is an example of *fuzzy logic*.
- In particular, causation is *retrospective* and need not involve *predetermination*.
- **Even if isolated events have predetermined causal structure, the causally closed universe need not be predetermined!**

  [They] may not think that the idea of a total state of the universe at a time is one [they] can do anything with. [They] may even have no views on the uniqueness of possible results for whatever may be going on in any arbitrary volume of space. For "Our theory should be such that only the actual result was possible for that experiment" doesn't mean "Our theory should have excluded the experimenter's being muffed or someone's throwing a boot, so that we didn't get the result", but rather: "Our theory should be such that only this result was possible as the result of the experiment". — G. E. M. Anscombe
- The issue here at hand is that given the state of the universe at some time, it could cause a **novel** future state, i.e. without predetermination. Worse yet, **there is no way to verify the effect of alternative initial states** as there is only one accessible universe, by definition.

Intentionality

- An **intentional system** is an agent which has 'mental' states whose contents refer to other mental states and/or objects external to the agent (Franz Bentano, 1838-1917).
- **Free will** is when an indeterministic agent behaves according to its intentional states.
- In *reductionist realism* for intentional states, there can be multiple physical correlates corresponding to an act of will, so free will does not obviously contradict statistical laws.
  However, owing to the novelty of intentionally evolving states, these statistical laws might be complex, if not impossible to draw inductive inferences from without updation of data.
- In *instrumentalism*, there are no physical states directly corresponding to intentional states and their relation is rather in subvenience, respectively. An example of such philosophy is that of Daniel Dennett (1942-2024).

Object ontology

- Recall that in our setting of Davidson's event ontology, events are what one ontologically commits to, whereas objects are replaced by statistical predicates.
- Consequently, stable objects (which are ideal) are replaced by a series of occurences of well-defined events, which maintain apparent stability through continuous causation.
- In general, events can be statistically collected together via roughly well-behaves statistical properties to do with the causal structure of the basic events.
- For example, let $\Omega \subseteq \prod_{\alpha \in f} A_\alpha$ be the **configuration space** of a physical system $\mathcal{S}$ with $|f|$ degrees of freedom. Ω encodes the positions and velocities of the said particles and causes the configuration to update to Ω' at a later time, with respect to the poset $\left(\prod_{\alpha \in f} A_\alpha, \preceq\right)$ generated by a timelike foliation on $\mathbb{R}^4$, and a probability space $\left(\prod_{\alpha \in f} A_\alpha, \mathcal{P}\left(\prod_{\alpha \in f} A_\alpha\right), \mathsf{pr}\right)$,

$$\Omega \rightsquigarrow \Omega'$$

Cluster predicates

- Due to the *fuzzy* nature of statistical inferences and often, imprecise causal models, causal inference is restricted to cluster predicates (i.e. predicates constrolled by creteria), as opposed to universals.
- Let W be the set of all **logically possible worlds**. Then, for each $w \in W$, there may be contingent conditions allowing us to infer $S_w(\Omega) \rightsquigarrow S_w(\Omega')$. S_w can reduce to different predicates depending on $w \in W$.
- This is **Hilary Putnam**'s (1926-2016) notion of *cluster concepts* in disguise[6]:

  I am inclined to say that 'pain' is a cluster-concept. That is, the application of the word 'pain' is controlled by a whole cluster of criteria, all of which can be regarded as synthetic. As a consequence, there is no satisfactory way of answering the question 'What does "pain" mean?' except by giving an exact synonym (e.g. 'Schmerz'); but there are a million and one different ways of saying what pain is. One can, for example, say that pain is that feeling which is normally evinced by saying 'ouch', or by wincing, or in a variety of other ways (or often not evinced at all). — Hilary Putnam

[6] Here, the causal relationship between pain and its contingent effects is analogous to that between initial and final configurations in event space.

- For example, in a logical world $w \in W$ where Ω roughly predetermines Ω', one can make all sorts of inferences of the form $S_w(\Omega) \rightsquigarrow S_w(\Omega')$. However, in some other logical world $w' \in W$, it is possible that $S_w(\Omega) \not\rightsquigarrow S_w(\Omega')$, so causal inference has to resort to other appropriate predicates, in the manner $S_{w'}(\Omega) \rightsquigarrow S_{w'}(\Omega')$.
- The fuziness involved in such inferences creates a gradation of object-driven to process-driven ontologies of objects:

$$\begin{tikzcd}
\text{Logical positivism} \arrow[r] & \text{Instrumentalism} \arrow[r] & \text{Process philosophy} \arrow[d] \\
 & & \text{Skepticism}
\end{tikzcd}$$

Figure: Fuzzy gradation of ontologies for physical objects.

Logical positivism: Frege

- **Gottlob Frege** (1848-1925) revolutionized mathematics, linguistics and philosophy in general, via his series of realizations in logic.
- His first realization we consider is that **sentences are more fundamental than subjects or predicates alone** when it comes to justifying their logical structure,

  A distinction of subject and predicate finds no place in my way of representing a judgement. In order to justify this, let me observe that there are two ways in which the content of two judgements may differ; it may, or may not, be the case that all inferences that can be drawn from the first judgement when combined with certain other ones can also be drawn from the second when combined with the same other judgements.
  — Gottlob Frege
- The second important idea of Frege's[7] is the **contextual principle**: "*never to ask for the meaning of a word in isolation, but only in the context of a proposition*".

[7] anticipated by Jeremy Bentham (1748-1832) in his notion of implicit definitions.

- The effect of contextuality in first-order logic is that there is not necessarily **an** isolated translation x' for a subject x such that $P(x') \iff P(x)$ for all relevant predicates P.
- Instead, depending on the predicate/context, x can have different translations, i.e. **contextual definitions**,

$$
\begin{aligned}
P(x) &\iff P(x') \\
Q(x) &\iff Q(x'') \\
&\vdots \\
x &\neq x'' \neq \cdots
\end{aligned}
$$

- The third important epiphany of Frege's is that of **two-factor semantics** of **sense and reference**:
  1. "*always to separate sharply the psychological from the logical, the subjective from the objective*"
  2. "*never to lose sight of the distinction between concept and object*"

- In Frege's theory of sense (intension) and reference (extension), subjects such as proper names have the following semantical aspects:
  1. A *mode of presentation* associated with the name of the subject, called the **sense/intension**.
  2. An *object* that the name of the subject talks about, i.e. the subject's **reference/extension**.
- Two-factor semantics solves many philosophical puzzles, such as in **identity theory**. Specifically, two subjects with different senses can refer to the same object, so they can simultaneously be presentationally non-identical but extensionally identical.
- For example, we could have objects $x = y$ (e.g. the evening star Hesperus and the morning star Phosphorus) such that to an agent S, in the context of a predicate P (such as '*is Venus*'), x and y have different modes of presentation. Then, x and y cannot be inter-substituted in propositional attitude statements concerning P,

$$B_S\left[Q\left(P(x), z, \ldots\right)\right] \not\Rightarrow B_S\left[Q\left(P(y), z, \ldots\right)\right]$$

- The referents of sentences are '*true*' and '*false*'. In an **extensional logic** (which only looks at referents of subjects), the referent of a sentence does not depend on the senses of its subjects and there are valid **substitution rules**.

Intuitionistic logic

- In **intensional logics**, logical entities have semantical structure deeper than their reference, so substitution is not always valid.
- For example, in lambda calculus, substitution only works at the α (hyperintensional) or β (extensional) levels, but not in between them,

$$\lambda x\,[(x+5)+7] \neq \lambda x\,[x+12]$$
$$\forall M : (\lambda x\,[(x+5)+7])\,M = (\lambda x\,[x+12])\,M$$

- In modal logics such as possible world semantics, contingent substitutivity of propositions, due to their contextual sense, does not extend to necessary or possible versions of the propositions.
  For example, let x be the class of people, P the predicate '*reads papers*' and S the predicate '*is a scientist*'.
  Then, $\forall x : S(x) \implies P(x)$. However, this is not logically necessary, so we **cannot perform the substition**:

$$\Box\,[\forall x : S(x) \implies P(x)]$$

  since there are logically possible worlds where being a scientist does not entail reading books (and say, absorbing information via other means).

- In general, intuitionistic logics drop the principles of *the excluded middle* and *double negation elimination*.
- The previous construction of causal structure is intuitionistic in the sense that $\neg\widehat{\text{cor}}\,(A, B)$ could mean $\widehat{\overline{\text{cor}}}\,(A, B)$ *or* $\text{cor}\,(A, B) = 0$, and likewise for causal relationships.
- In Goldman-esque causal theories of knowledge, the sense corresponding to an empirical subject, in the context of a predicate, can depend on the causal relationship of the agent and the referent of the subject.

  Therefore, sentences can refer to not only '*true*' and '*false*' but also an additional state to the effect of '*indeterminate*'.
- The above view is formalized in **topos theory**, which is a higher-order intuitionistic logic. In topoi, not only is the degree of truth a fundamental logical entity, but so is a notion of *where* a proposition is truth-evaluable.[8]

[8] '*Topos*' in Greek means '*place*'.

Logical positivism: Russell

- Bertrand Russell's mathematical *magnum opus* is his work with Alfred North Whitehead, the ***Principia Mathematica***, published 1910-1913 (PM). In it:
  1. There is a first-order deductive logic, manifesting as a formal system for manipulating symbols.
  2. Set theory is a *logical construction* out of logic.
  3. Similarly, number theory is a logical construction out of set theory and so on.
- Therefore, an aspiration of PM-like systems is to provide a **translation** for a theory (such as number theory) built logically atop a more fundamental theory (here, formal logic and set theory).

  This is a **deductive foundationalist** approach to structured theories.
- Logical positivism is the extreme anti-metaphysical extension of the above framework to supposed double knowledge theories. If there are two such theories, the logical positivist holds that they co-refer to the empirical contents of either one of them, or those of a more fundamental foundational theory (Herbert Feigl, 1902-1988) — in spite of their different senses as theories.

Identity theory

- Logical positivism also employs the **verification criterion** for the meaning of sentences in a theory; non-empirical statements are not truth-evaluable. This is why logical positivism is anti-metaphysical.
- A common pattern in the translation of the empirical contents of one theory to that of another is **identification** of objects. The semantics of identification is reflected in that of the word '*is*', which, according to Russell, is of 4 types:
  1. The '*is*' of **identity**: e.g. *A cat* is *a cat*.
  2. The '*is*' of **equality**: e.g. *Hesperus* is *Phosphorus*.
  3. The '*is*' of **predication**: e.g. *FemPhys* is *cool*.
  4. The '*is*' of **composition**: e.g. *Lightning* is *electric discharge*.
- U. T. Place (1924-2000) categorizes the above kinds of '*is*' into two groups:
  1. Those that do not admit the '*and nothing else*' rider: identity, equality
  2. Those that admit the '*and nothing else*' rider: predication, composition

  Furthermore, in the latter category, only composition is of contingent nature, whereas predication, by virtue of involving Platonic concepts, makes definitions universal.

Compositional identity

- Place would say that '*lightning is electric discharge*' is not an *a priori* true or false statement, as compositional identity is contingent. Moreover, the contextual independence of lightning-talk (such as its phenomenology) and electric-discharge-talk (electrodynamics) does not imply their ontological independence — such would only be the case if in a *common* context, lightning and electric discharge talks are distinct.
- In fact, lightning-talk refers to electric-discharge-talk and this direction of reduction is the aim of a logical positivist's enterprise of translation in a common context.
- Since compositional identity is contenigent, statements about them are truth-evaluable by the methods of science. In fact, non-verificational claims about such statements are semantically meaningless in logical positivism.

Synthesis

- Putting it all together, the following picture emerges in classical materialism, if we wish to ask such questions as '*what is real in a theory of electrons?*' (notice the transition and non-exclusivity of the below views):

1. **Rationalism**: *An electron is a beable whose nature is partly empirical and partly deduced via reasoning with falsifiable consequences on observables.* Therefore, an electron is a singular type of entity which should rationally explain, for instance, the logical positivist's contextual definitions of an electron and the event ontologist's descriptions of electron-like events.
2. **Logical positivism**: *An electron is the referent of contextual empirical statements in a scientific theory of electrons that translates experimental-talk to electron-talk.*
   For example, in the context of a scattering experiment, an electron refers to that beable which produces observations such as electron cloud photographs.
   In another context such as Millikan's oil drop experiment, 'electron' translates to 'that beable which in the setting of electrodynamics explains the behaviour of certain oil droplets'.

3. **Object ontology**: *An electron is at best a 'fuzzy' beable; in statistically normal conditions, what it means to be 'an electron' reduces to certain predicates; however, in non-normal conditions, such predicates are controlled by other criteria. In other words, 'is an electron' is a contingent cluster predicate with no logical translations.*
4. **Event ontology**: *There are only electron-like events with statistically significant causal relationships, gleaned from experimental interaction.*
5. **Instrumentalism**: *The above conceptions of an electron are useful in different theoretical and experimental settings; whether non-empirical, metaphysical statements are true is irrelevent to the scientific process.*
6. **Empiricism**: *Like in rationalism, an electron is the referent of a rational construct that explains scientific data. Unlike in rationalism, such rationalization can be completely learnt via exposure to sense data; there need not be rational principles that exist independently of our empirical lives, in the construction of scientific theories.*
7. **Transcendental idealism**: *An electron may refer to a physical object committed to in the above views; however, the electron is uncognizable as a thing-to-itself and there must be a transcendental agent with synthetic a priori cognitive constructs to have meaningful electron-talk.*

8. **Objective idealism**: *There are no material objects. Instead, there is a mind-independent, neutral, mentalistic Substance that represents a total Sum of causes and provides conceptual structure to reality. Parts of the Sum are partial reflections of the whole (parallelism) and the working of the mentalistic substance can habitually change (tychism) continuously (synechism).*
   When agents perceive the world, this is a one-sided and partial expression of the Substance, and multiple such expressions capture some essence of the Sum (pluralism).
9. **Subjective idealism**: *Like in objective idealism, there are no material objects with stable mind-independent existence; unlike in objective idealism, existence of observables is solely by virtue of perception of a transcendent agent and not via any mind-independent substance.*
10. **Skepticism**: *There is no logical justification for beliefs originating in inductive reasoning on empirical data or the contents of theories. Unlike in rationalism and like in empiricism, rational principles cannot provide such logical justification as they are learned from sense data in the first place.*
11. **Coherentism**: *A wide range of views to the effect of foundationalism and reductionist physicalism being unjustifiable. Instead, various propositions with cognitive and empirical content co-justify each other and which ones are foundational depends on the context and level of locality in a knowledge superstructure.*